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METHOD FOR ALIGNING BACKSIDE PATTERN BASED ON ONE OR MORE FRONTSIDE ALIGNMENT MARKS OF A SEMICONDUCTOR WAFER

Abstract

A method comprises the steps of providing a semiconductor wafer comprising a substrate layer; forming a first alignment mark and a second alignment mark; forming a plurality of super junction buffer regions; forming additional two or more epitaxial layers; forming additional one or more alignment marks; forming a top layer; thinning the wafer; forming a first plurality of regions and a second plurality of regions; and applying a singulation process. Devices made by the method have reduced misalignment, between a top portion and a bottom portion of an interface within the devices, for less than 0.5 micron.

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Background/Summary

FIELD OF THE INVENTION

[0001] This invention relates generally to a method of fabricating semiconductor devices by precisely aligning backside pattern to frontside alignment marks of a semiconductor wafer. More particularly, the present invention relates the method referencing alignment marks distanced from a circumference of the semiconductor wafer.

BACKGROUND OF THE INVENTION

[0002] Fabrication of a semiconductor device containing a super junction metal-oxide-semiconductor field-effect transistor (SJ-MOSFET) and a reverse conducting insulated-gate bipolar transistor (RC-IGBT) requires N and P alternating patterns on a backside of a silicon wafer. The backside pattern is not precisely aligned with a frontside pattern because the silicon wafer faces down (the frontside pattern is not accessible) during a backside patterning process. A conventional fabrication method using a notch on the silicon wafer introduces large alignment errors (in the order of 30 microns) between the backside pattern and the frontside pattern.

[0003] The present disclosure references one or more frontside alignment marks distanced from a circumference of the semiconductor wafer. The alignment accuracy is improved so that the alignment error is in the order of 0.5 micron.

SUMMARY OF THE INVENTION

[0004] The present invention discloses a method for fabricating semiconductor devices. The method comprises the steps of providing a semiconductor wafer comprising a substrate layer; forming a first alignment mark and a second alignment mark; forming a plurality of super junction buffer regions; forming additional two or more epitaxial layers; forming additional one or more alignment marks; forming a top layer; thinning the wafer; forming a first plurality of regions and a second plurality of regions; and applying a singulation process.

[0005] Conventionally, MOSFET used N+ substrate and IGBT used P- substrate. In examples of present disclosure, RC-IGBT N-Substrate is used as a common substrate for the MOSFET and IGBT cells so that the SJ-MOSFET and the RC-IGBT are integrated side-by-side.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a flowchart of a process to fabricate a plurality of semiconductor devices in examples of the present disclosure.

[0007] FIG. 2A is a top view and FIGS. 2B, 2C, 2D, 2E, 2F, 2G, 2H, 2I, 2J, 2K, 2L, and 2M show cross sectional views of steps of the process to fabricate the plurality of semiconductor devices of FIG. 1 in examples of the present disclosure.

[0008] FIGS. 3A and 3B are top views of two different configurations of alignment marks in examples of the present disclosure.

[0009] FIG. 4 shows a semiconductor device in examples of the present disclosure.

[0010] FIG. 5 shows light beams capturing images of one or more alignment marks in examples of the present disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0011] FIG. 1 is a flowchart of a process **100** to fabricate a plurality of semiconductor devices in examples of the present disclosure. The process **100** may begin in block **102**.

[0012] In block **102**, referring now to FIGS. 2A and 2B, a semiconductor wafer **202** is provided. The semiconductor wafer **202** comprises a substrate layer **204**. In one example, the substrate layer **204** is made of silicon having 8-inch diameter and 725 um thickness. In another example, the substrate layer **204** is made of silicon having 12-inch diameter and 775 um thickness. In examples of the present disclosure, the substrate layer **204** is a doped first conductivity type silicon substrate layer. In one example, the first conductivity type is N type. The substrate layer comprises a first side **206** and a second side **208**. The second side **208** opposites the first side **206**. In examples of the present disclosure, the substrate layer **204** further comprises a cutout **209**. FIGS. 2B-2M are viewed toward the cross-sectional plane AA'. Block **102** may be followed by block **104**.

[0013] In block **104**, referring now to FIG. 2B, an epitaxial layer **212** is formed on the first side **206** of the substrate layer **204**. In one example, the epitaxial layer **212** is lightly doped with N type. The epitaxial layer **212** comprises a first side **216** and a second side **218**. The second side **218** opposites the first side **216**. The second side **218** of the epitaxial layer **212** is attached to the first side **206** of the substrate layer **204**. Block **104** may be followed by block **106**.

[0014] In block **106**, referring now to FIGS. 2C and 3A, a first alignment mark **222** and a second alignment mark **224** are formed on the first side **216** of the epitaxial layer **212**. Block **106** may be followed by block **108**.

[0015] In block **108**, referring now to FIG. 2D, a photoresist pattern **232** is formed on the first side **216** of the epitaxial layer **212**. Though FIGS. 2D-2M show the process to fabricate three devices, the number of devices in a cross-sectional plane may vary. Block **108** may be followed by block **110**.

[0016] In block **110**, referring now to FIG. 2E, a plurality of super junction buffer regions **242** are formed on the first side **216** of the epitaxial layer **212**. In one example, the super junction buffer regions **242** are doped with N type. Block **110** may be followed by block **112**.

[0017] In block **112**, referring now to FIG. 2F, the photoresist pattern **232** of FIG. 2E is removed so as to expose the surface **249** of the first side **216** of the epitaxial layer **212**. Block **112** may be followed by block **114**.

[0018] In block **114**, referring now to FIG. 2G, additional two or more epitaxial layers **252** are formed on top of the epitaxial layer **212**. The additional two or more epitaxial layers **252** comprises a first side **251** and a second side **253** opposite the first side **251**. The second side **253** of the additional two or more epitaxial layers **252** is attached to the first side **216** of the epitaxial layer **212**. In one example, a thickness of each layer of the additional two or more epitaxial layers **252** is 6 microns. Though six epitaxial layers (**253A**, **253B**, **253C**, **253D**, **253E**, and **253F**) are shown in FIG. 2G, the number of epitaxial layers may vary. P and/or N type dopants are implanted into each epitaxial layer to form doped P columns **421** and doped N columns **423** shown in FIG. 4 in each device area.

[0019] In examples of the present disclosure, the number of epitaxial layers is an even number. In examples of the present disclosure, intermediate alignment marks are formed every two epitaxial layers. In one example, the additional two or more epitaxial layers **252** consists of two additional epitaxial layers. There are no intermediate alignment marks. In another example, the additional two or more epitaxial layers **252** consists of four additional epitaxial layers. There are two intermediate alignment marks. In still another example, the additional two or more epitaxial layers **252** consists of six additional epitaxial layers including a first additional epitaxial layer **253A**, a second additional epitaxial layer **253B**, a third additional epitaxial layer **253C**, a fourth additional epitaxial layer **253D**, a fifth additional epitaxial layer **253E**, and a sixth additional epitaxial layer **253F**. There are four intermediate alignment marks including a first lower intermediate alignment mark **255A**, a second lower intermediate alignment mark **257A**, a first higher intermediate alignment

mark **255B**, and a second higher intermediate alignment mark **257B**.

[0020] The first lower intermediate alignment mark **255A** is formed between the second additional epitaxial layer **253B** and the third additional epitaxial layer **253C**. The second lower intermediate alignment mark **257A** is formed between the second additional epitaxial layer **253B** and the third additional epitaxial layer **253C**. The first higher intermediate alignment mark **255B** is formed between the fourth additional epitaxial layer **253D** and the fifth additional epitaxial layer **253E**. The second higher intermediate alignment mark **257B** is formed between the fourth additional epitaxial layer **253D** and the fifth additional epitaxial layer **253E**.

[0021] In block **116**, referring now to FIGS. **2H** and **3A**, a first refreshed alignment mark **262**, a second refreshed alignment mark **264**, and additional one or more alignment marks **266** are formed on the first side **251** of the additional two or more epitaxial layers **252**. In one example, a distance between a center **351** of FIG. **3A** of the wafer and a first one **266A** of the additional one or more alignment marks **266** is in a range from 20% to 70% of a radius of the wafer. In another example, a distance between a center **351** of FIG. **3A** of the wafer and the first alignment mark **222** is larger than 90% of the radius of the wafer.

[0022] In one example, referring now to FIGS. **2H** and **3A**, the additional one or more alignment marks **266** comprise a single alignment mark. In one example, referring now to FIGS. **2H** and **3B**, the additional one or more alignment marks **266** comprise four alignment marks **362**, **364**, **366**, and **368**. Each of the four alignment marks **362**, **364**, **366**, and **368** is positioned at a respective vertex of a square shape. In examples of the present disclosure, for a 6-inch (152.4 mm) diameter wafer or a 8-inch (203.2 mm) diameter wafer, the coordinates of the four alignment marks **362**, **364**, **366**, and **368** are (44.5 mm, 48.3 mm), (−44.5 mm, 48.3 mm), (−44.5 mm, −48.3 mm), and (44.5 mm, −48.3 mm) respectively (origin of the coordinate is at the center of the top surface of the wafer). In examples of the present disclosure, a length of the side of the square shape is 65.67 mm. Block **116** may be followed by block **118**.

[0023] In block **118**, referring now to FIGS. **21** and **4**, a top layer **272** is formed on the first side **251** of the additional two or more epitaxial layers **252**. A SJ-MOSFET and an RC-IGBT top structures are formed in each device area of the top layer **272** using the refreshed alignment marks **262**, **264** and the additional one or more alignment marks **266** as reference. The top layer **272** comprises a plurality of source/emitter and body regions (for example, a plurality of source/emitter regions **412** and body regions **413** of semiconductor device **400** of FIG. **4**) and a plurality of gate regions (for example, a gate region **432** of semiconductor device **400** of FIG. **4**). A top metal layer may be deposited over the top layer **272** to form the device top electrodes. Block **118** may be followed by block **120**.

[0024] In block **120**, referring now to FIG. **2J**, the substrate layer of FIG. **21** is removed so as to expose the surface **219** of epitaxial layer **212**. Block **120** may be followed by block **122**.

[0025] In block **122**, referring now to FIG. **2K**, a grinding process is applied to the surface **219** at the second side **218** of the epitaxial layer **212** so as to form a thinned epitaxial layer **213** comprising an exposed surface **282**. Block **122** may be followed by block **124**.

[0026] In block **124**, referring now to FIG. **2L**, the wafer is flipped so that the thinned epitaxial layer **213** is on top for processing. A first plurality of regions **284** and a second plurality of regions **286** are formed on the exposed surface **282** of the thinned epitaxial layer **213** by referencing the additional one or more alignment marks **266**. In one example, the first plurality of regions **284** and the second plurality of regions **286** are doped with opposite conductivities. Back metal layer may be deposited onto the exposed surface **282** to form the bottom electrode of the device.

[0027] FIG. **5** shows light beams capturing images of one or more alignment marks in examples of the present disclosure. In one example, the tool **502** is an ASML 3DALIGN™ product described in paragraph of US Patent Application Publication US2019/0006285 to Zhang et al. The tool **502** emits light beam **512A**. Light beam **512A** is reflected by mirror **532** becoming light beam **512B**. Light beam **512B** is reflected by mirror **534** becoming light beam **512C**. The returning light beam

is shown in dashed lines. Light beam **514A** is reflected by mirror **534** becoming light beam **514B**. Light beam **514B** is reflected by mirror **532** becoming light beam **514C**. An image of the additional one or more alignment marks **266** is captured on a focal plane **599**. The first plurality of regions **584** and the second plurality of regions **586** are aligned using the image. Therefore, referring now to FIG. **4**, a respective misalignment between a top portion **492T** and a bottom portion **492B** of an interface **492** between the respective SJ-MOSFET **402** and the respective RC-IGBT **452** is less than 0.5 micron (reduced from 30 microns).

[0028] In examples of the present disclosure, block **124** comprises the sub-steps of forming the first plurality of regions **284** being a plurality of doped second conductivity type regions; forming an additional photoresist pattern **289** (shown in dashed-lines because of being removed in a later sub-step); forming the second plurality of regions **286** being a plurality of heavily doped first conductivity type regions; and removing the additional photoresist pattern **289**. In examples of the present disclosure, the plurality of doped second conductivity type regions are a plurality of doped P type regions. The plurality of heavily doped first conductivity type regions

[0029] are a plurality of heavily doped N type regions. Paragraph of US patent Application Publication No. US2019/0148165 to He et al. recites that “In examples of the present disclosure, heavily doped has ion concentration in a range above 10^{18} cm.^{sup.}−3. Doped has ion concentration in a range from 10^{16} to 10^{18} cm.^{sup.}−3. Lightly doped has ion concentration in a range below 10^{16} cm.^{sup.}−3.” Block **124** may be followed by block **126**.

[0030] In block **126**, referring now to FIG. **2M**, the wafer is flipped again so that the top layer **272** is on top. A singulation process, comprising a plurality of cutting lines **291**, is applied so as to form the plurality of semiconductor devices **292**. Though only three semiconductor devices **292A**, **292B**, and **292C** are shown in FIG. **2M**, the number of semiconductor devices fabricated from a wafer may vary.

[0031] FIG. **4** illustrates an example for a semiconductor device **400** fabricated by the method **100**. The semiconductor device **400** comprises a SJ-MOSFET **402** area and an RC-IGBT **452** area. The SJ-MOSFET **402** and the RC-IGBT **452** each comprise a plurality of source/emitter regions **412**, a plurality of body regions **413**, and a plurality of gate regions **432** on the top portion of the device. A plurality of P columns **421** and an N columns **423** form a super junction structure in the middle portion of the device. The SJ-MOSFET **402** area further comprise a drain region **482** under a second buffer region **494** and the RC-IGBT **452** area further comprises a collector region **483** under a first buffer region **493**. The second buffer region **494** may be the super junction buffer regions **242** shown in FIG. **2F** and the first buffer region **493** may be the epitaxial layer **212**. The drain region **482** of the SJ-MOSFET and the collector region **483** of the RC-IGBT have opposite conductivities. Metal layers on the top and bottom surfaces of the device **400** are not shown.

[0032] Those of ordinary skill in the art may recognize that modifications of the embodiments disclosed herein are possible. For example, a number of layers of the additional two or more epitaxial layers may vary. Other modifications may occur to those of ordinary skill in this art, and all such modifications are deemed to fall within the purview of the present invention, as defined by the claims.

Claims

1. A method for fabricating a plurality of semiconductor devices, the method comprising the steps of: providing a semiconductor wafer comprising: a substrate layer being a doped first conductivity type silicon substrate layer, the substrate layer comprising a first side and a second side opposite the first side; forming an epitaxial layer on the first side of the substrate layer, the epitaxial layer comprising a first side and a second side opposite the first side, the second side of the epitaxial layer being attached to the first side of the substrate layer; forming a first alignment mark and a second alignment mark on the first side of the epitaxial layer; forming a photoresist pattern on the

first side of the epitaxial layer; forming a plurality of super junction buffer regions on the first side of the epitaxial layer; removing the photoresist pattern; forming additional two or more epitaxial layers, the additional two or more epitaxial layers comprising a first side and a second side opposite the first side, the second side of the additional two or more epitaxial layers being attached to the first side of the epitaxial layer; forming a first refreshed alignment mark, a second refreshed alignment mark, and additional one or more alignment marks on the first side of the additional two or more epitaxial layers, a distance between a center of the wafer and a first one of the additional one or more alignment marks is in a range from twenty percent to seventy percent of a radius of the wafer; forming a top layer; removing the substrate layer; applying a grinding process to the second side of the epitaxial layer forming a thinned epitaxial layer comprising an exposed surface; forming a first plurality of regions and a second plurality of regions on the exposed surface of the thinned epitaxial layer by referencing the additional one or more alignment marks; and applying a singulation process forming the plurality of semiconductor devices.

2. The method of claim 1, wherein the additional one or more alignment marks comprise a single alignment mark.

3. The method of claim 1, wherein the additional one or more alignment marks comprise four alignment marks; and wherein each of the four alignment marks is positioned at a respective vertex of a square shape.

4. The method of claim 1, wherein the step of forming the first plurality of regions and the second plurality of regions comprises the sub-steps of: forming the first plurality of regions being a plurality of doped second conductivity type regions; forming an additional photoresist pattern; forming the second plurality of regions being a plurality of heavily doped first conductivity type regions; and removing the additional photoresist pattern.

5. The method of claim 4, wherein the doped first conductivity type silicon substrate layer is a doped N type silicon substrate layer; wherein the plurality of doped second conductivity type regions are a plurality of doped P type regions; and wherein the plurality of heavily doped first conductivity type regions are a plurality of heavily doped N type regions.

6. The method of claim 5, wherein the top layer comprises: a plurality of source regions; and a plurality of gate regions.

7. The method of claim 1, wherein each device of the plurality of semiconductor devices comprises: a respective super junction metal-oxide-semiconductor field-effect transistor (SJ-MOSFET); and a respective reverse conducting insulated-gate bipolar transistor (RC-IGBT).

8. The method of claim 7, wherein a respective misalignment between a top portion and a bottom portion of an interface between the respective SJ-MOSFET and the respective RC-IGBT is less than one-half of a micron.

9. The method of claim 1, the step of forming the first plurality of regions and the second plurality of regions on the exposed surface of the thinned epitaxial layer by referencing the additional one or more alignment marks comprising the sub-steps of capturing an image of the additional one or more alignment marks on a focal plane; and aligning the first plurality of regions and the second plurality of regions using the image.

10. The method of claim 1, wherein the additional two or more epitaxial layers consists of: a first additional epitaxial layer; and a second additional epitaxial layer.

11. The method of claim 1, wherein the additional two or more epitaxial layers consists of: a first additional epitaxial layer; a second additional epitaxial layer; a third additional epitaxial layer; and a fourth additional epitaxial layer; wherein during the step of forming the additional two or more epitaxial layers, a first intermediate alignment mark is formed between the second additional epitaxial layer and the third additional epitaxial layer; and a second intermediate alignment mark is formed between the second additional epitaxial layer and the third additional epitaxial layer.

12. The method of claim 1, wherein the additional two or more epitaxial layers consists of: a first additional epitaxial layer; a second additional epitaxial layer; a third additional epitaxial layer; a

fourth additional epitaxial layer; a fifth additional epitaxial layer; and a sixth additional epitaxial layer; wherein during the step of forming the additional two or more epitaxial layers, a first lower intermediate alignment mark is formed between the second additional epitaxial layer and the third additional epitaxial layer; a second lower intermediate alignment mark is formed between the second additional epitaxial layer and the third additional epitaxial layer; a first higher intermediate alignment mark is formed between the fourth additional epitaxial layer and the fifth additional epitaxial layer; and a second higher intermediate alignment mark is formed between the fourth additional epitaxial layer and the fifth additional epitaxial layer.

13. A semiconductor device formed on a semiconductor chip, the semiconductor device comprising: a super junction metal-oxide-semiconductor field-effect transistor (SJ-MOSFET) area; and a reverse conducting insulated-gate bipolar transistor (RC-IGBT) area; wherein the SJ-MOSFET area comprises: a plurality of source regions, a plurality of body regions, a plurality of gate regions on a top portion of the SJ-MOSFET area, and a drain region on a bottom portion of the SJ-MOSFET area; wherein the RC-IGBT area comprises: a plurality of emitter regions, a plurality of body regions, a plurality of gate regions on a top portion of the RC-IGBT area, and a collector region on a bottom portion of the RC-IGBT area; and wherein a respective misalignment between a top portion and a bottom portion of an interface between the SJ-MOSFET area and the RC-IGBT area is less than 0.5 micron.

14. The semiconductor device of claim 13 further comprising a plurality of P columns and a plurality of N columns forming a super junction structure in a middle portion of the semiconductor device.

15. The semiconductor device of claim 13 further comprising a super junction buffer region above the drain region of the SJ-MOSFET area.
